

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building event-driven AI infrastructure, distributed systems, and full-stack analytics platforms. Scaled an agentic AI orchestration platform to 2,000 async inference requests/second using Temporal workflow orchestration and Kafka; implemented low-level virtual memory systems in C at the kernel level; shipped 10+ backend features across 7 production deployments at an AI-native startup. Experienced collaborating with research teams on ML architecture and model evaluation pipelines. Seeking to contribute to frontier AI infrastructure and training systems.

CORE COMPETENCIES

Event-Driven AI Infrastructure

Distributed Systems (Temporal + Kafka)

Full-Stack Engineering (Next.js / TypeScript)

Low-Level Systems Programming (C / Kernel)

AI Agent Orchestration

Analytics Pipelines & Observability

Multi-Model LLM Pipelines

Docker / Kubernetes / CI-CD

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Built **Temporal.io workflow orchestration** infrastructure for an AI-powered claims automation platform, managing multi-step agentic pipelines across 3 AI providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models with fault-tolerant state machines
- Designed and shipped **document processing pipelines** (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and real-time event pipelines using Redis and Pusher - 7 production deployments with zero regressions
- Implemented role-based access control, multi-model AI agent tooling with MCP, and Auth0 authentication; contributed **394 commits** across 10+ backend features in TypeScript, Node.js, Express, PostgreSQL, and Prisma
- Integrated Sentry observability and Playwright for automated testing; participated in full PR review cycle on a 2-person engineering team

NC State University April 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Designed a **cloud-based genomic data infrastructure** platform with JBrowse 2, enabling interactive exploration of 500GB strawberry and grape datasets for 100+ researchers worldwide, deployed on Google Cloud Run
- Collaborated directly with ML research team on **model architecture design**, applying hyperparameter tuning and time-series analysis pipelines - improved model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern - Remote

- Built 5+ **AI-driven automation workflows** including a health claim automation tool using OCR, NLP, and ML pipelines; improved claim accuracy by 25%
- Automated infrastructure deployments on AWS and GCP; integrated **monitoring dashboards with Grafana**; supported model validation and red-teaming to enhance reliability

Capgemini Jun 2023 - Aug 2023

AI/ML Intern - Remote

- Led end-to-end development of a **distributed data pipeline** for support ticket analysis using Hadoop + Spark for batch processing and Kafka streaming for real-time ingestion - reduced ticket-processing latency by 20%
- Reduced overall processing time by 15% through pipeline optimization and automated NLP-based classification

PROJECTS

AlgoPulse - Agentic AI Orchestration Platform

AI Infrastructure

Architected an **event-driven AI orchestration platform** for async inference at scale. Identified an API-polling bottleneck under load and redesigned the architecture around Supabase Edge Functions streaming AI reasoning loops directly to clients. Scaled worker infrastructure with **Temporal.io and Kafka for state synchronization and fault tolerance** - 2,000 async inference requests/second without state loss.

Stack: TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka | [github.com/Aditya-k24/AlgoPulse](#)

Demand Paging - Xinu OS Kernel

Low-Level Systems

Engineered an **8-store virtual memory system in C** for Xinu OS, implementing Second-Chance page replacement, inverted page tables, and a custom fault handler. Added 7+ kernel system calls and a new demand paging infrastructure - reduced page faults by 25% and improved memory performance by 15%.

Stack: C, Xinu OS, x86 virtual memory, kernel systems programming

PulseRank - Real-Time Leaderboard & Activity Feed

Full-Stack + Analytics

Built an **event-driven analytics infrastructure**: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for feed document projections with TTL expiry, **Redis sorted sets for sub-millisecond leaderboard reads**, live score updates over Server-Sent Events. Shipped full observability with OpenTelemetry + Prometheus; benchmarked hot paths with k6.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, OpenTelemetry, Prometheus | [github.com/Aditya-k24/PulseRank](#)

LLM Eval Harness - Vendor-Neutral Benchmark

AI Evaluation Pipelines

Built a **model evaluation and data pipeline** comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000 frozen test examples. Tracked ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, and cost/call. Applied statistical significance via paired bootstrap CIs, McNemar test, and Holm correction.

Stack: Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker | [github.com/Aditya-k24/llm-eval-harness](#)

EDUCATION

MS Computer Science - North Carolina State University

Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science - Mumbai University

Aug 2020 - May 2024

Mumbai, India | CGPA: 9.32/10 | Courses: Big Data, Machine Learning, DS & Algorithms, Blockchain

SKILLS

Languages: Python, TypeScript, Java, JavaScript, C, R, SQL

AI/ML: LLM APIs (Anthropic, OpenAI, Gemini), MCP, LangChain, RAG, TensorFlow, PyTorch, Hugging Face

Infrastructure: Temporal.io, Kafka, Docker, Kubernetes, AWS, GCP, CI/CD, Grafana, OpenTelemetry, Prometheus

Full-Stack: Node.js, React, Next.js, NestJS, PostgreSQL, MySQL, MongoDB, Redis, Prisma, Express

Systems: Spark, Hadoop, Distributed Systems, Kernel Programming (C), Virtual Memory, Beam